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HTTP Request API

Hint

While this API is not deprecated, it is highly recommended to use the JSON API instead of the HTTP API for new integrations, as it is structured in a better way and allows efficient use of newer features like segments, presets, and playlists.

WLED's HTTP API allows you to set many properties of your lights, even more than the index page UI supports, via a simple GET web request.

Help

Unsure how all this API stuff works? Check out this amazing guide by tynick!

The basic URL scheme is: `[ipaddress]/win`. This will return an XML file with some current values (see bottom of page). Parameters can be added to control some of the variables.

- Example (AP): `192.168.4.1/win&A=255` sets the brightness to maximum
- Example (mdns): `led.local/win&A=128&FX=0` sets the brightness to half and the effect to Solid

In conjunction with a router port forwarding this can be used to automate WLED, for example via IFTTT.

Add one or multiple of the following parameters after the base URL/IP to change values: (if the parameter is unknown or the value illegal nothing will happen)

LED control

Parameter	Value Range	Description	Since Version
&A=	0 to 255	Master brightness	0.2
&T=	0, 1, or 2	Master Off/On/Toggle	0.3
&R=	0 to 255	Primary Red value	0.2
&G=	0 to 255	Primary Green value	0.2
&B=	0 to 255	Primary Blue value	0.2
&W=	0 to 255	Primary White value	0.4
&FX=	0 to 101	LED Effect Index	0.3
&SX=	0 to 255	Effect Speed	0.3
&IX=	0 to 255	Effect Intensity	0.5.0
&FP=	0 to 46	FastLED Palette	0.8.0
&NL=	0 to 255	Nightlight active and duration in minutes	0.3
&ND	none	Toggles nightlight on but uses default duration	0.6.3
&NT=	0 to 255	Nightlight target brightness	0.5.0
&NF=	0 to 2	Fade Nightlight, 1 = fade brightness only, 2 = additionaly fade color from primary to secondary color	0.5.0

Advanced

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Parameter	Value Range	Description	Since Version
&CL=	HEX/DEC	Primary color	0.8.0
&C2=	HEX/DEC	Secondary color	0.8.0
&C3=	HEX/DEC	Third color	0.8.0
&R2=	0 to 255	Secondary Red value	0.4
&G2=	0 to 255	Secondary Green value	0.4
&B2=	0 to 255	Secondary Blue value	0.4
&W2=	0 to 255	Secondary White value	0.4
&HU=	0 to 65535	Hue	0.5.1
&SA=	0 to 255	Saturation (only in conjunction with Hue)	0.5.1
&H2	none	Hue + Saturation will set secondary color	0.5.1
&SR=	0 or 1	Set Primary/Secondary color to random hue	0.4
&SC	none	Swap primary and secondary color	0.4

Use of hex values

Hex values need to be prefaced with the character h or H. The normal format is `RRGGBB`. If the led strip is RGBW, the hex format is `WWRRGGBB`. Note: In the UI the format is `RRGGBBWW`, so the values cannot be copied without a transformation.

Loxone commands

Loxone offers two commands. One for RGB values and one for brightness and color temperature.

Parameter	Syntax	Range	Description	Since Version
&LX=	BBBGGGRRR	0 - 100100100	Loxone RGB value for primary color. Each color (`RRR`, `GGG`, `BBB`) is specified in the range from 0 to 100%.	0.11
&LX=	20bbbtttt	200002700 - 201006500	Loxone brightness and color temperature values for primary color. Brightness `bbb` is specified in the range 0 to 100%. `tttt` defines the color temperature in the range from 2700 to 6500 Kelvin.	0.11
&LY=	BBBGGGRRR	0 - 100100100	Loxone RGB value for secondary color. Each color (`RRR`, `GGG`, `BBB`) is specified in the range from 0 to 100%.	0.11
&LY=	20bbbtttt	200002700 - 201006500	Loxone brightness and color temperature values for secondary color. Brightness `bbb` is specified in the range 0 to 100%. `tttt` defines the color temperature in the range from 2700 to 6500 Kelvin.	0.11

Notifications

Parameter	Value Range	Description	Since Version
&RN=	0 or 1	Receive UDP Notifications	0.3
&SN=	0 or 1	Send UDP Notifications	0.3
&NN	none	No notification for this request	0.3
&HP=	0 to 99	Sets Hue polling light ID (0 is off)	0.5.1

Presets

Parameter	Value Range	Description	Since Version
&PS=	1 to 16	Saves current setup to preset. Preset 255 can be used and is temporary/not persistent.	0.4
&PL=	1 to 250	Applies entire preset	0.4
&P1=	1 to 249	First cycle preset	0.6.3
&P2=	2 to 250	Last cycle preset	0.6.3
&TT=	0 to 65000	Set transition time (ms)	0.6.3

Macros

Parameter	Value Range	Description	Since Version
&M=	1 to 16	Apply macro (deprecated, added for compatibility with pre-0.11 automations)	0.5.0

Segments

It is highly recommended to use the JSON API when dealing with Segments.

Parameter	Value Range	Description	Since Version
&SM=	0 to 9	Set the main segment (values are reported to XML)	0.9.0
&SS=	0 to 9	Select segment to apply THIS api call to	0.9.0
&SV=	0, 1, or 2	Set segment selected (2 unselects others)	0.9.0
&S=	0 to ledcount-1	Set segment start	0.9.0
&S2=	0 to ledcount	Set segment stop	0.9.0
&GP=	1 to 255	Set segment grouping	0.9.1
&SP=	0 to 255	Set segment spacing	0.9.1
&RV=	0 or 1	Reverse/Flip Segment direction	0.9.1
&SB=	0 to 255	Segment brightness	0.10.0

General and Experimental

Parameter	Value Range	Description	Since Version
&RB	none	Reboot WLED	0.10.0 (?)
&ST=	32bit	Current UTC time in Unix epoch	0.4
&CT=	32bit	UTC time for countdown end	0.4
&MD=	0 or 1	Set slider mode to RGB/HSB	0.3
&AX=	0 to 255	Debug feature, can be configured for general IO	0.3
&IN	none	Server will not respond to this request (internal)	0.3
&OL=	0 to 255	Experimental overlays	0.3
&L=	0 to 255	Lock pixel	0.4
&L2=	0 to 255	Lock pixel range L to L2	0.4
&UL	none	Unlock instead (used in conjunction with L and L2)	0.4
&NX=	String 1..6	Cronixie clockface	0.4
&NM=	0 or 1	Cronixie Time or Countdown mode	0.4
&NB=	0 or 1	Cronixie Backlight	0.4
&IT	none	Include UI color theme in API response	0.8.2
&RD=	0 or 1	Toggle realtime UDP	0.8.4
&LO=	0-2	Live data override. 0 is off, 1 is override until live data ends, 2 is override until ESP reboot	0.10.2
&NP	none	Advance to the next preset in a playlist	0.15

XML response

This is the XML file sent as response to every API call.

Parameter	Value Range	Description
ac	0 to 255	Master Brightness
cs	3x 0..255	Secondary RGB
ns	0 or 1	Notification Sending on
nr	0 or 1	Notification Receive on
nl	0 or 1	Nightlight active
nf	0 or 2	Nightlight Fade type
nd	0 to 255	Nightlight delay
nt	0 to 255	Nightlight target brightness
fx	0 to 73	Effect index
sx	0 to 255	Effect speed
ix	0 to 255	Effect intensity
fp	0 to 43	FastLED palette
wv	-1 to 255	Primary White value
ws	0 to 255	Secondary White
ps	0 to 255	Current Preset
cy	0 or 1	Preset Cycling enabled
md	0 or 1	RGB or HSB UI mode
ds	String 0..32	Server description
ss	0 to 12	Segment ID

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In-/decrementing values

You can use the `~` character to easily set values relative to their current value.
This is currently supported for the following parameters:
`A, R, G, B, W, R2, G2, B2, W2, FX, SX, IX, FP, PL`

For example, use `PL=~` to go to the next preset. Using just `~` without a number will increase the value by 1, `~-` will decrease it by 1. The value will then wrap around, so using `A=~-` when A is 0 will set A to 255.

You can also specify by how much to change the value. For example, using `A=~10` will increase the brightness by 10. In case of using a number behind `~`, the value will clip (so it will not wrap around, if the maximum brightness is set, `A=~10` will not have any effect)

To setup a Macro for a Button to advance to the next Preset, use **win&P1=1&P2=30&PL=~**
P1 will equal the first Preset of the rotation while P2 will be the last Preset.